QUASI-FINITENESS AT POINTS OVER A ONE-DIMENSIONAL LOCAL RING FROM CONDITIONS AT GENERIC POINTS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file varieties.tex; no statement or proof step was rewritten, reordered or removed.

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Editorial note. The preamble is the chapter's retained minimal preamble (sources/stacks-varieties/retained-preamble.tex lines 1-77), a verbatim excerpt of the native preamble.

Editorial note. The label lemma-quasi-finite-in-codim-1 is defined only in varieties.tex, so no cross-chapter restatement had to be weighed for this card.

Allowed standard prerequisites (editorial).

- Quasi-finite locus of a locally of finite type morphism is open. Morphisms, Lemma morphisms-lemma-quasi-finite-points-open, invoked at proof lines 3211-3212. Supplies that a dense set of points at which f is quasi-finite contains a dense open, hence all generic points of irreducible components.
- Quasi-finiteness at a point versus finiteness of the residue field extension. Morphisms, Lemma morphisms-lemma-residue-field-quasi-finite and Lemma morphisms-lemma-quasi-finite-at-point-characterize, invoked at proof lines 3215-3216, 3235-3236 and 3263-3264. Supplies the translation between the hypotheses at generic points and the conclusion at fibre points.
- Spec of a local ring as the set of points specializing to the point, and base change of quasi-finiteness. Schemes, Lemma schemes-lemma-specialize-points and Morphisms, Lemma morphisms-lemma-base-change-quasi-finite, invoked at proof lines 3221-3225. Supplies the reduction that lets Y be assumed to be the spectrum of a Noetherian local ring of dimension at most one.
- Dimension formula for a morphism of schemes. Morphisms, Lemma morphisms-lemma-dimension-formula, invoked at proof lines 3246-3247. Supplies the inequality from which the vanishing of the transcendence degree is read off.

Source reference. Original lemma 01TE: sources/stacks-chow/tags.txt, source lines 2371–2371. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Date: stacks-project:varieties.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Source reference. Original lemma 01TM: sources/stacks-chow/tags.txt, source lines 2379–2379. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 02JU: sources/stacks-chow/tags.txt, source lines 3349–3349. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 01T6: sources/stacks-chow/tags.txt, source lines 2363–2363. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 01TH: sources/stacks-chow/tags.txt, source lines 2374–2374. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 01TI: sources/stacks-chow/tags.txt, source lines 2375–2375. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 01TG: sources/stacks-chow/tags.txt, source lines 2373–2373. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 01J7: sources/stacks-chow/tags.txt, source lines 2004–2004. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Lemma 0.1. *Let $f : X \to Y$ be locally of finite type. Let $y \in Y$ be a point such that $\mathcal{O}_{Y,y}$ is Noetherian of dimension ≤ 1. Assume in addition one of the following conditions is satisfied*

1. *for every generic point η of an irreducible component of X the field extension $\kappa(\eta)/\kappa(f(\eta))$ is finite (or algebraic),*
2. *for every generic point η of an irreducible component of X such that $f(\eta) \rightsquigarrow y$ the field extension $\kappa(\eta)/\kappa(f(\eta))$ is finite (or algebraic),*
3. *f is quasi-finite at every generic point of an irreducible component of X,*
4. *Y is locally Noetherian and f is quasi-finite at a dense set of points of X,*
5. *add more here.*

Then f is quasi-finite at every point of X lying over y.

Proof. Condition (4) implies X is locally Noetherian (Morphisms, Lemma 01T6). The set of points at which morphism is quasi-finite is open (Morphisms, Lemma 01TI). A dense open of a locally Noetherian scheme contains all generic point of irreducible components, hence (4) implies (3). Condition (3) implies condition (1) by Morphisms, Lemma 01TG. Condition (1) implies condition (2). Thus it suffices to prove the lemma in case (2) holds.

Assume (2) holds. Recall that $\mathrm{Spec}(\mathcal{O}_{Y,y})$ is the set of points of Y specializing to y, see Schemes, Lemma 01J7. Combined with Morphisms, Lemma 01TM this shows

we may replace Y by $\mathrm{Spec}(\mathcal{O}_{Y,y})$. Thus we may assume $Y = \mathrm{Spec}(B)$ where B is a Noetherian local ring of dimension ≤ 1 and y is the closed point.

Let $X = \bigcup X_i$ be the irreducible components of X viewed as reduced closed subschemes. If we can show each fibre $X_{i,y}$ is a discrete space, then $X_y = \bigcup X_{i,y}$ is discrete as well and we conclude that $X \to Y$ is quasi-finite at all points of X_y by Morphisms, Lemma 01TH. Thus we may assume X is an integral scheme.

If $X \to Y$ maps the generic point η of X to y, then X is the spectrum of a finite extension of $\kappa(y)$ and the result is true. Assume that X maps η to a point corresponding to a minimal prime $\mathfrak{q}$ of B different from $\mathfrak{m}_B$. We obtain a factorization $X \to \mathrm{Spec}(B/\mathfrak{q}) \to \mathrm{Spec}(B)$. Let $x \in X$ be a point lying over y. By the dimension formula (Morphisms, Lemma 02JU) we have

$$\dim(\mathcal{O}_{X,x}) \leq \dim(B/\mathfrak{q}) + \mathrm{trdeg}_{\kappa(\mathfrak{q})}(R(X)) - \mathrm{trdeg}_{\kappa(y)}\kappa(x)$$

We know that $\dim(B/\mathfrak{q}) = 1$, that the generic point of X is not equal to x and specializes to x and that $R(X)$ is algebraic over $\kappa(\mathfrak{q})$. Thus we get

$$1 \leq 1 - \mathrm{trdeg}_{\kappa(y)}\kappa(x)$$

Hence every point x of X_y is closed in X_y by Morphisms, Lemma 01TE and hence $X \to Y$ is quasi-finite at every point x of X_y by Morphisms, Lemma 01TH (which also implies that X_y is a discrete topological space). $\square$